Data Preprocessing, Feature Engineering & Machine Learning Pipeline Development

Dataset: IBM HR Analytics Employee Attrition Dataset

1. Dataset Overview

The dataset was loaded and inspected using Python, Pandas and NumPy. The original dataset contained 1,470 records and 35 columns. Attrition was used as the target variable for the machine learning workflow.

2. Data Quality Assessment

Missing values detected: **0**
Duplicate records detected: **0**

The dataset was checked for missing values, duplicate records, data types and categorical values before preprocessing.

3. Data Cleaning

The following columns were removed because they were constant or identifier-related and were not useful as predictive features:

Column	Reason
EmployeeCount	Constant value
EmployeeNumber	Employee identifier
Over18	Constant value
StandardHours	Constant value

4. Feature Engineering

CompanyExperienceRatio: Created by dividing YearsAtCompany by TotalWorkingYears. Zero was used when TotalWorkingYears was zero to avoid division by zero.

CareerStage: TotalWorkingYears was grouped into Early Career, Developing, Experienced and Senior categories.

5. Outlier Detection

Potential outliers were identified using the Interquartile Range (IQR) method and visualized using box plots. Age, MonthlyIncome, TotalWorkingYears and YearsAtCompany were examined. Extreme observations were reviewed rather than automatically deleted because they may represent legitimate employee observations.

6. Categorical Encoding

One-Hot Encoding was used in the final preprocessing pipeline for categorical features. The encoder was configured with handle_unknown='ignore'. Label Encoding was also demonstrated separately for comparison.

7. Feature Scaling

StandardScaler was used in the final numerical preprocessing pipeline. MinMaxScaler was also applied to MonthlyIncome as a comparison exercise. StandardScaler standardizes features around a mean of approximately

zero, whereas MinMaxScaler maps values to a bounded range.

8. Machine Learning Pipeline

A ColumnTransformer was created with separate numerical and categorical preprocessing workflows. Numerical features use median imputation followed by StandardScaler. Categorical features use most-frequent imputation followed by One-Hot Encoding.

The data was divided into training and testing sets using an 80:20 ratio, random_state=42 and stratification based on Attrition. The preprocessing pipeline was fitted only on the training data and then applied to the testing data to reduce data leakage.

9. Saved Outputs

The preprocessing pipeline was saved as **preprocessor.pkl**. The cleaned dataset was saved as **employee_attrition_clean.csv**. The implementation was saved as **preprocessing_pipeline.py**.

10. Assumptions and Limitations

The preprocessing workflow was designed for preparation of the Employee Attrition dataset for subsequent machine learning. Preprocessing choices may need to be adjusted for other datasets or production environments. Potential outliers were reviewed rather than universally removed.